

# Lecture 10

## Register Basics and Memory Unit

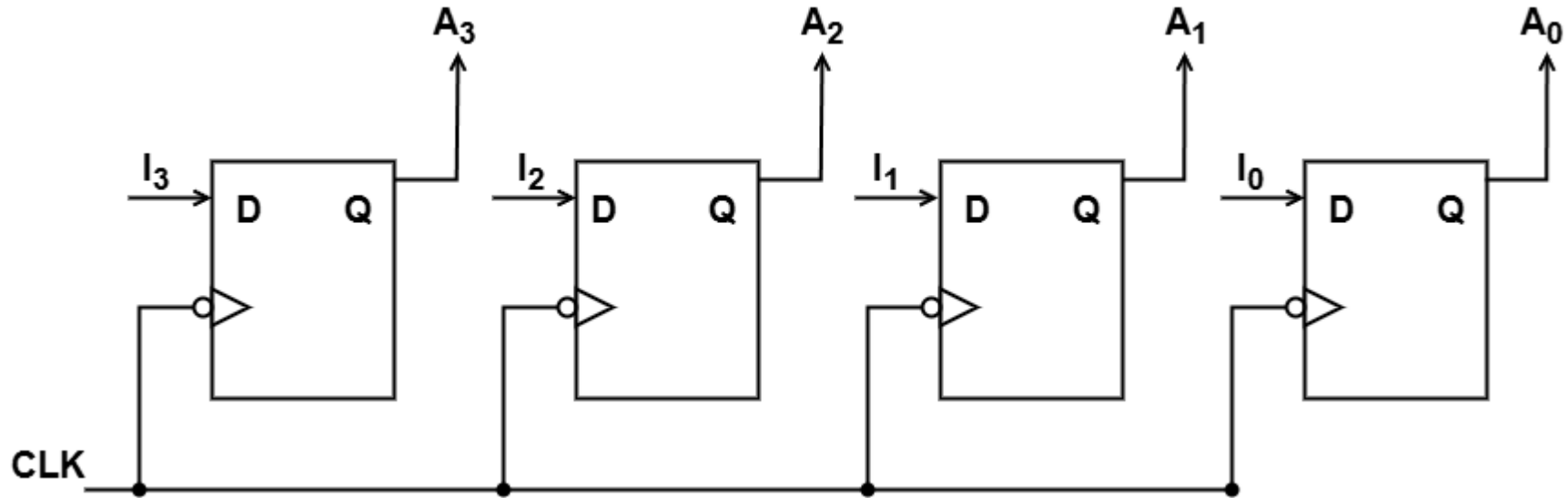


CSE260 : Digital Logic Design

# Registers

- Flip-flops can only store 1-bit of data
- Multiple flip-flops can be combined into a single group which is called a register
- A register can store multiple bits
- An n-bit register is consist of n flip-flops and different logic gates, it can store n-bit information
- The flip-flops in a register store information
- The logic gates and other components control the insertion and the flow of the data
- Functions of register
  - Storing data
  - Retrieving data from other registers
  - Shifting data within the register (right/left)

# Simple Register



- Example: A simple 4-bit register, consist of D flip-flops which does not contain any logic gate or other components. New 4-bit data is stored with every clock cycle.

# What is Memory Unit?

A **memory unit** is a **collection of registers together** with associated circuits needed to transfer information in and out of registers.

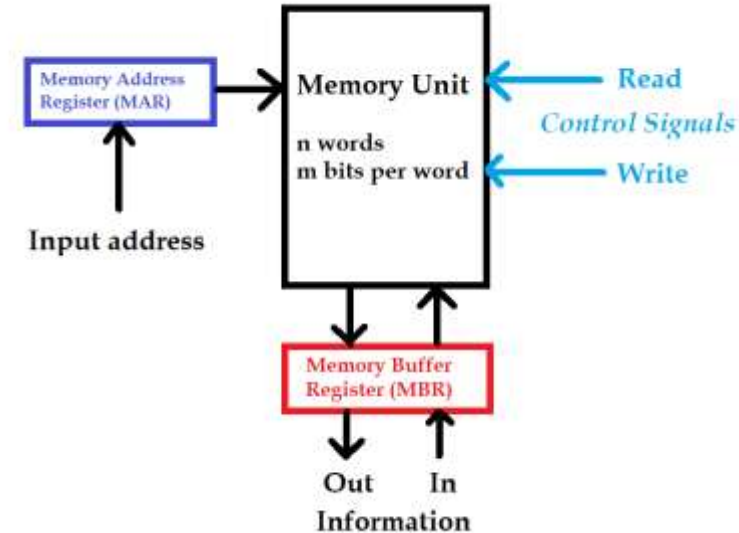
**Binary cell:** An elementary unit of **computer storage** (i.e. memory register) that can have one or the other of **two stable states** and can thus **store one bit of information**.

The memory cell is an electronic circuit that stores one bit of binary information and it must be set to store a logic 1 (high voltage level) and reset to store a logic 0 (low voltage level). Its value is maintained/stored until it is changed by the set/reset process.

# What is Memory Unit?

A memory unit stores binary information in groups of bits called *words*.

Communication between memory unit and its environment is achieved through **2 controls**(read and write) and **2 external registers** (memory address register and memory buffer register).

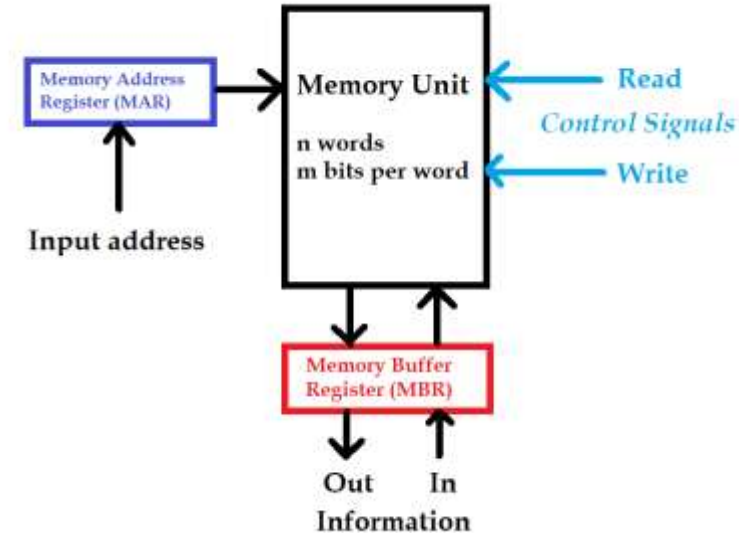


# What is Memory Unit?

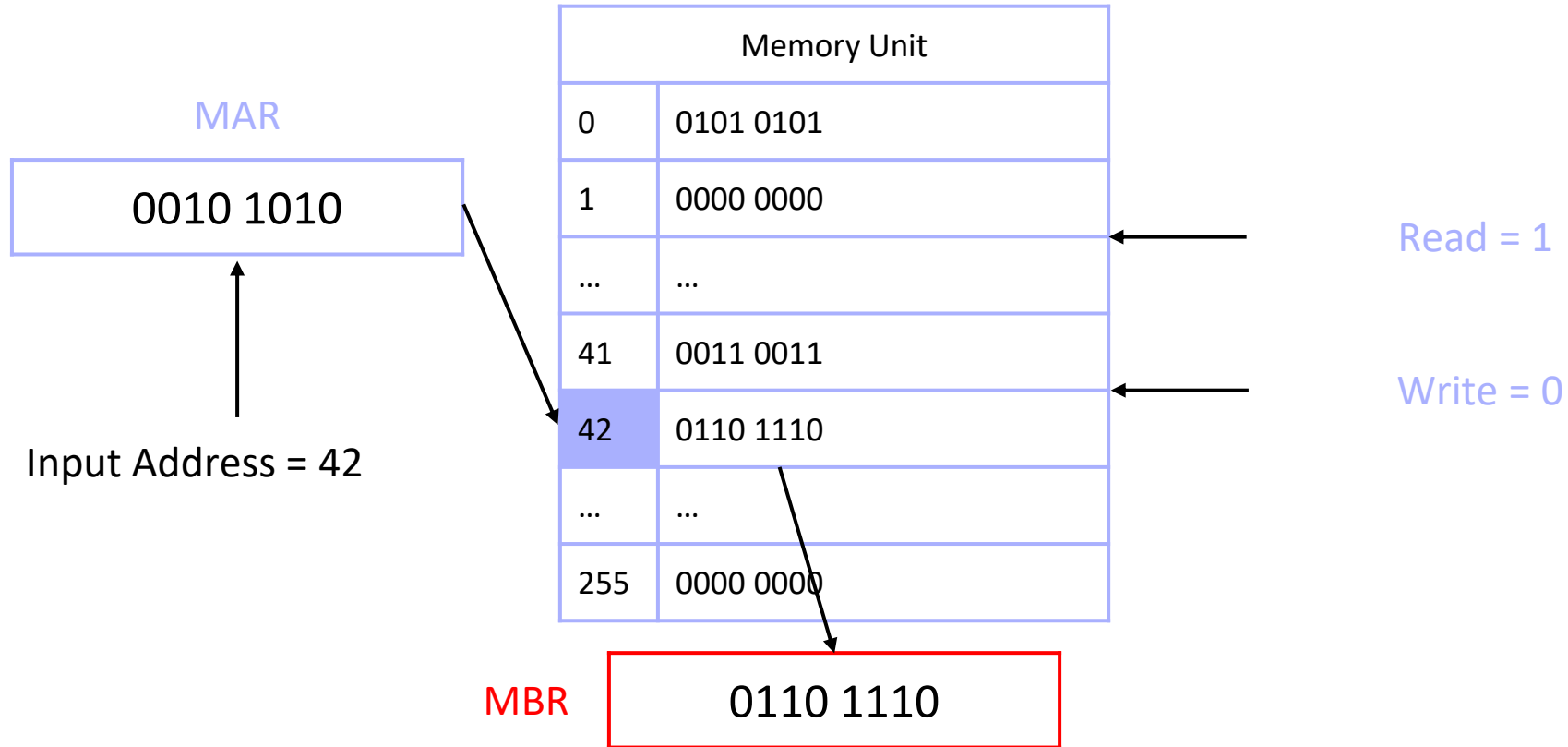
To communicate with a memory word, its address is transferred to **memory address register**.

Internal circuit then opens the path to select that word. A **n**-bit address register can specify up to  $2^n$  address. (note: # of address=# of word)

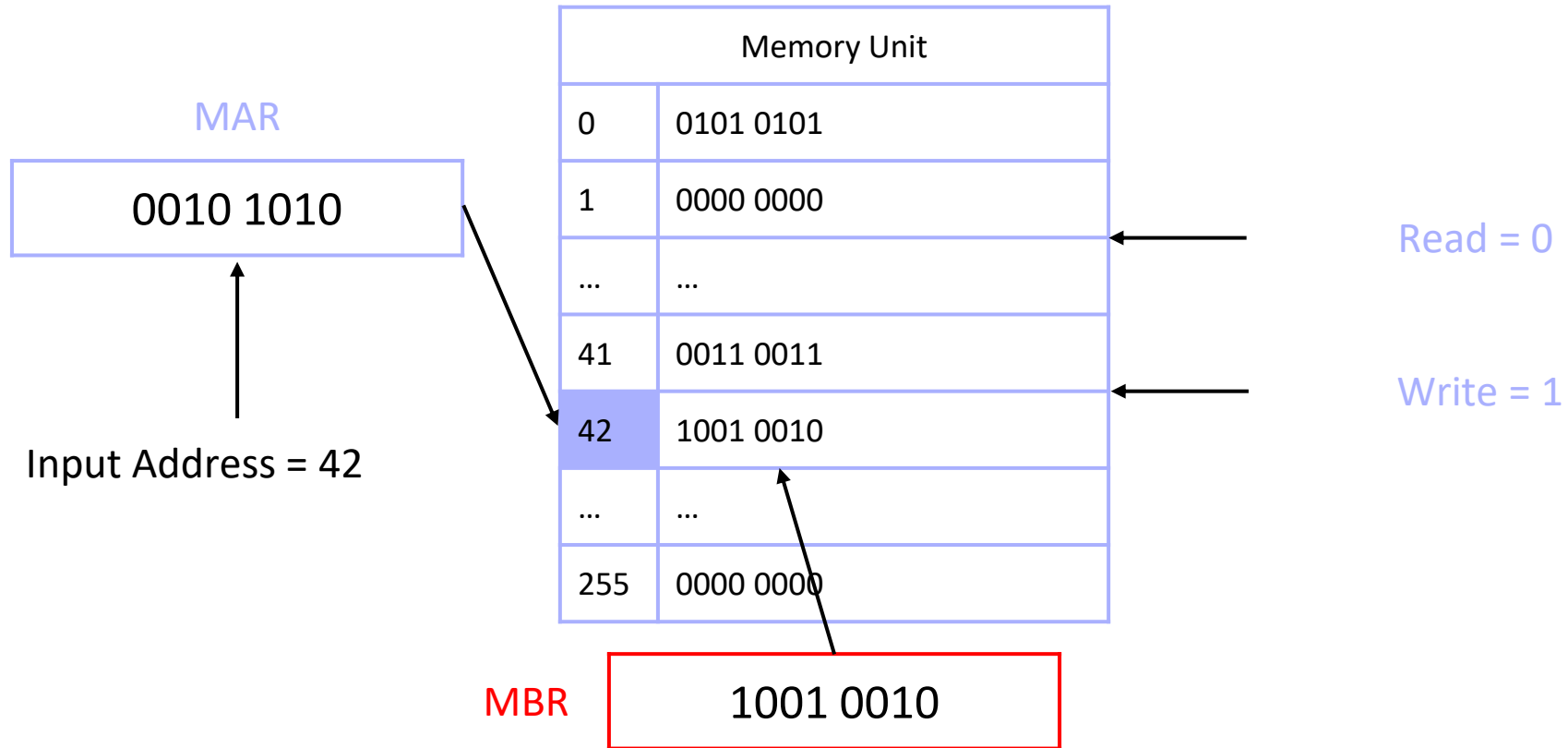
Before writing and after reading information are kept in **Memory Buffer register**. In each case, the address register specifies location of writing or reading.



# READ Operation Example



# WRITE Operation Example



# Content of a Memory

Content of a 1024 x 16-bit memory:

| Memory address |         | Memory content   |
|----------------|---------|------------------|
| binary         | decimal |                  |
| 0000000000     | 0       | 1011010111011101 |
| 0000000001     | 1       | 1010000110000110 |
| 0000000010     | 2       | 0010011101110001 |
| :              | :       | :                |
| :              | :       | :                |
| 1111111101     | 1021    | 1110010101010010 |
| 1111111110     | 1022    | 0011111010101110 |
| 1111111111     | 1023    | 1011000110010101 |

# Types of Semiconductor Memories

**ROM – Read Only Memory** - a type of memory that cannot be written, can only be read. Contents determined during the manufacture time

## **RAM – Random Access Memory**

Memory that can be both read and written during normal operation.

- ◆ Contents are volatile, i.e. will be lost on power off.
- ◆ Two types of RAM:
  - Static RAM (SRAM)
  - Dynamic RAM (DRAM)

# Random Access Memory (RAM)

The data consists of  $n$  lines (for  $n$ -bit words). **Data input lines** provide the information to be stored (*written*) into the memory, while **data output lines** carry the information out (*read*) from the memory.

The address consists of  $k$  lines which specify which word (among the  $2^k$  words available) to be selected for reading or writing.

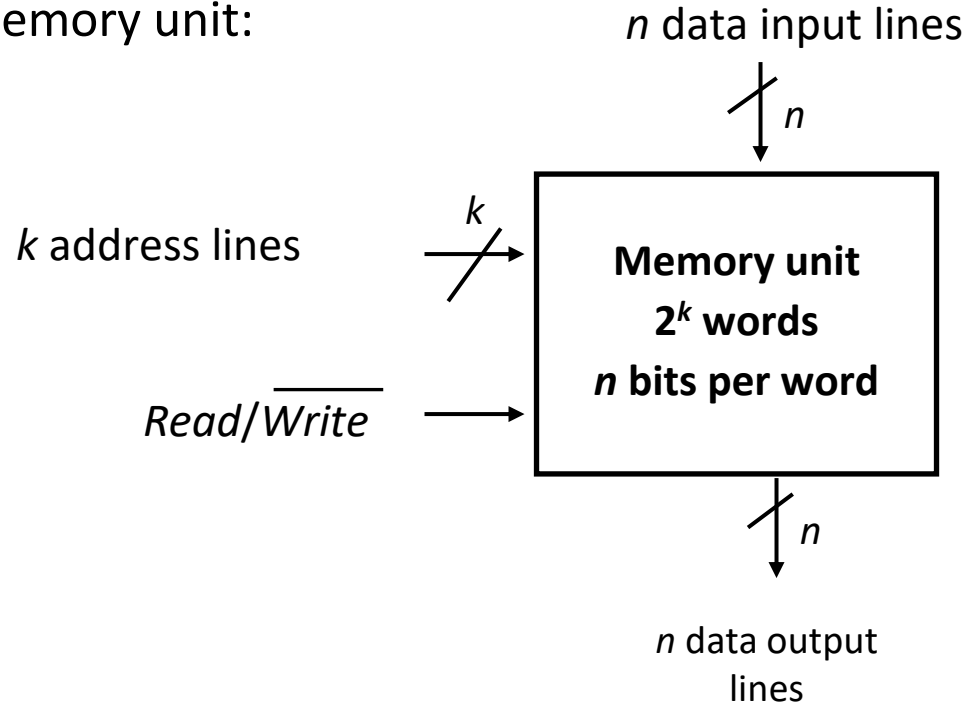
# Random Access Memory (RAM)

The control lines *Read* and *Write* (usually combined into a single control line *Read/Write*) specifies the direction of transfer of the data.

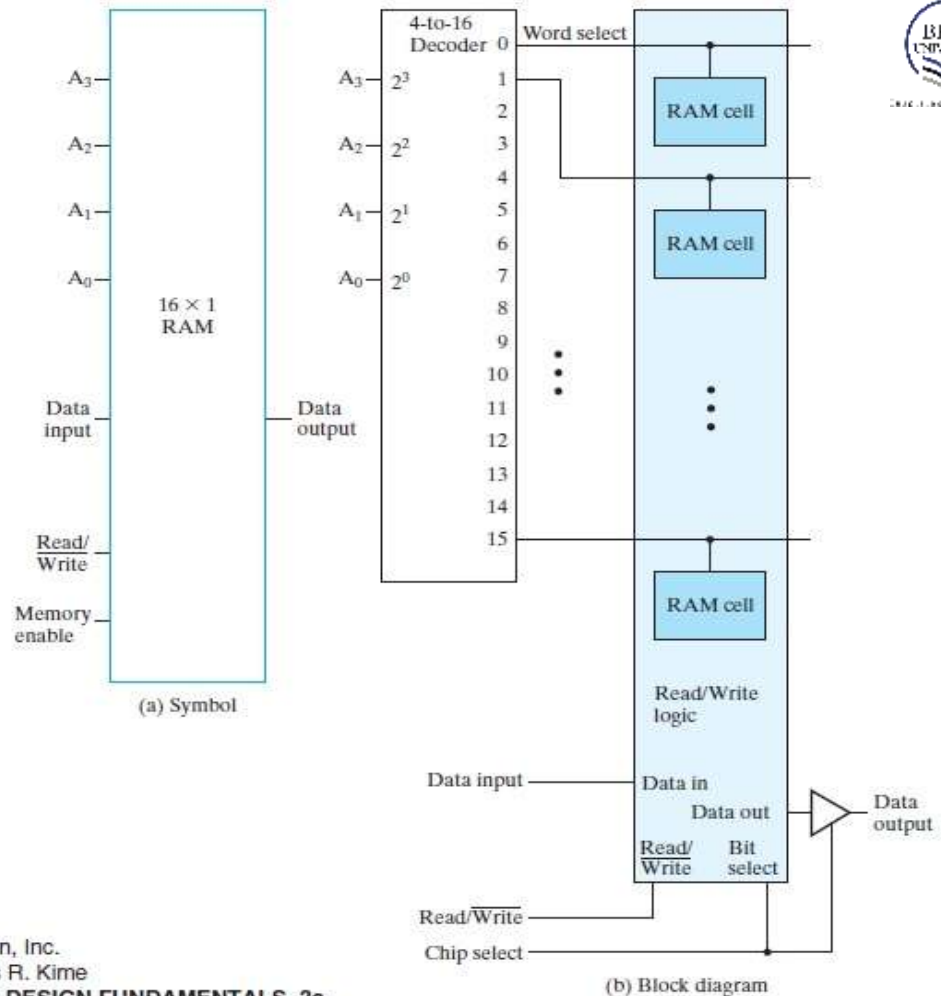
Example: A memory unit of 1024 words with 8 bit/word needs, 10 address line (i.e. address register must contain 10 FF) and buffer register must have 8 FF to store words transferred in and out of it.

# Random Access Memory (RAM)

Block diagram of a memory unit:



# Block Diagram of 16x1 RAM [no need to memorize]



# Random Access Memory (RAM)

The **Write** operation:

- ❖ Transfers the address of the desired word to the address lines
- ❖ Transfers the data bits (the word) to be stored in memory to the data input lines
- ❖ Activates the *Write* control line (set *Read/Write* to  $\overline{0}$ )

The **Read** operation:

- ❖ Transfers the address of the desired word to the address lines
- ❖ Activates the *Read* control line (set *Read/Write* to 1)

# Random Access Memory (RAM)

The Read/Write operation:

| Memory Enable | Read/Write | Memory Operation        |
|---------------|------------|-------------------------|
| 0             | X          | None                    |
| 1             | 0          | Write to selected word  |
| 1             | 1          | Read from selected word |

# Two types of RAM

## Static RAM (**SRAM**):

- Uses flip-flops as the memory cells, Complex Circuitry.
- No need to refresh constantly
- Typically the type of RAM used primarily as **cache memory**.

## Dynamic RAM (**DRAM**):

- Use capacitor charges to represent datas.
- Though simpler in circuitry, they have to be constantly refreshed.
- Most **main memory** is dynamic RAM

# Two types of RAM [Cont.]

## Dynamic RAM:

- **SDRAM** – Synchronous dynamic RAM – operates synchronously with system clock and data bus. Can handle 100MHz to >800MHz.
- **DDR – Double Data Rate** – can transmit data on both edges of the clock.  
**DDR3 = Double Data Rate 3rd generation.**
- **QDR – Quad Data Rate** – twice as fast as DDR

Typically we use DDR3, DDR4 & DDR5 RAM as main memory in our windows PC and Static RAM for L1, L2 and L3 cache memory.

# Memory Size

Memory sizes might be specified in bits or byte(=8bit)

Symbol: Bytes (B) and bits(b)

Example,

$$\begin{aligned}
 &2^{28} \text{ b} \\
 &= 2^{28}/8 \text{ B} \\
 &= 2^{28}/2^3 \text{ B} \\
 &= 2^{25} \text{ B} \\
 &= 2^5 * 2^{20} \text{ B} \\
 &= 32 \text{ MB}
 \end{aligned}$$

|   | Prefix | Base 2                   | Base 10              |
|---|--------|--------------------------|----------------------|
| K | Kilo   | $2^{10} = 1024$          | $10^3=1,000$         |
| M | Mega   | $2^{20} = 1,048,576$     | $10^6=1,000,000$     |
| G | Giga   | $2^{30} = 1,073,741,824$ | $10^9=1,000,000,000$ |

1 Byte = 8 Bits  
 1 KB = 1024 Bytes  
 1 MB = 1024 KB  
 1 GB = 1024 MB  
 1 TB = 1024 GB  
 1 PB = 1024 TB  
 1 EB = 1024 PB

# 2 formulas to remember

*Capacity = no. of word x bit/word*

*No. of word =  $2^{\text{address lines}}$*

Therefore,

*Capacity =  $2^{\text{address lines}}$  x bit/word*

# Test yourself

Capacity = no. of word x bit/word

No. of word =  $2^{\text{address lines}}$

What is the total capacity of a  $2^{16} \times 16$  memory?

# Test yourself [Sol<sup>n</sup>]

What is the total capacity of a  $2^{16} \times 16$  memory:

Solution:

$$= 2^{16} \times 16 \text{ bits/words}$$

$$= 1048576 \text{ bits}$$

$$= 1048576/8 \text{ bytes}$$

$$= 131072 \text{ bytes}$$

$$= 131072/1024 \text{ KB}$$

$$= 128 \text{ KB}$$

# Test yourself

Capacity = no. of word x bit/word

No. of word =  $2^{\text{(address lines)}}$

How many address lines do we need for a 64 MB RAM with 32 bit words?

# Test yourself [Sol<sup>n</sup>]

How many address lines do we need for a 64 MB RAM with 32 bit words:

Solution:

Formula:

Capacity = no. of word x bit/word

64MB = no. of word x 32 bits/word

no. of word = 64MB / 32 bits/word

$$= (64 \times 1024 \times 1024 \times 8) / 32 \quad [1 \text{ byte} = 2^3 \text{ bits}]$$

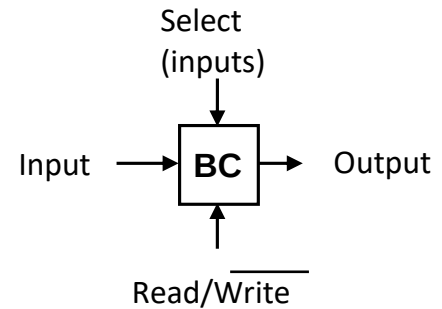
$$= 16777216$$

$$= 2^{24} \quad [\log_2(16777216) = 24]$$

no. of address lines = 24

# Binary cell

- Must be small so that many binary cell (BC) can fit in a small area
- Select input enable cell for reading/writing
- Read/write determines operation to be performed on cell. A **1 in read/write** input forms a path to the output terminal. A **0 in read/write input**, information in the input terminal is transferred into the FF

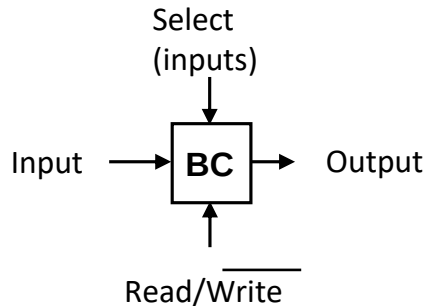


# Random Access Memory (RAM)

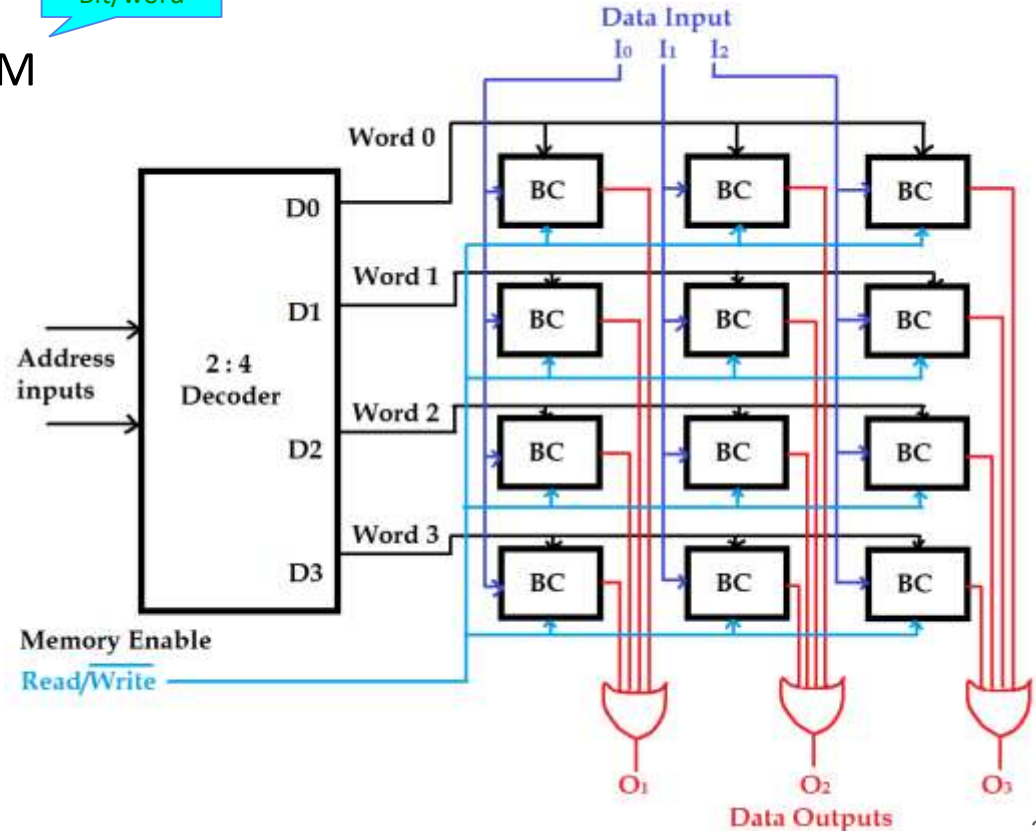
# of word

Bit/word

- Logic construction of a 4 x 3 RAM (with decoder and OR gates):



Since no of word is 4, that would be equivalent to the decoder's output. Since, the decoder will have 4 outputs, so we should use a 2x4 decoder

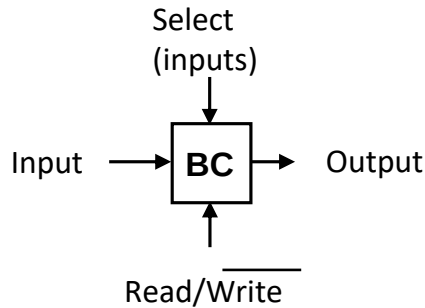


# Random Access Memory (RAM)

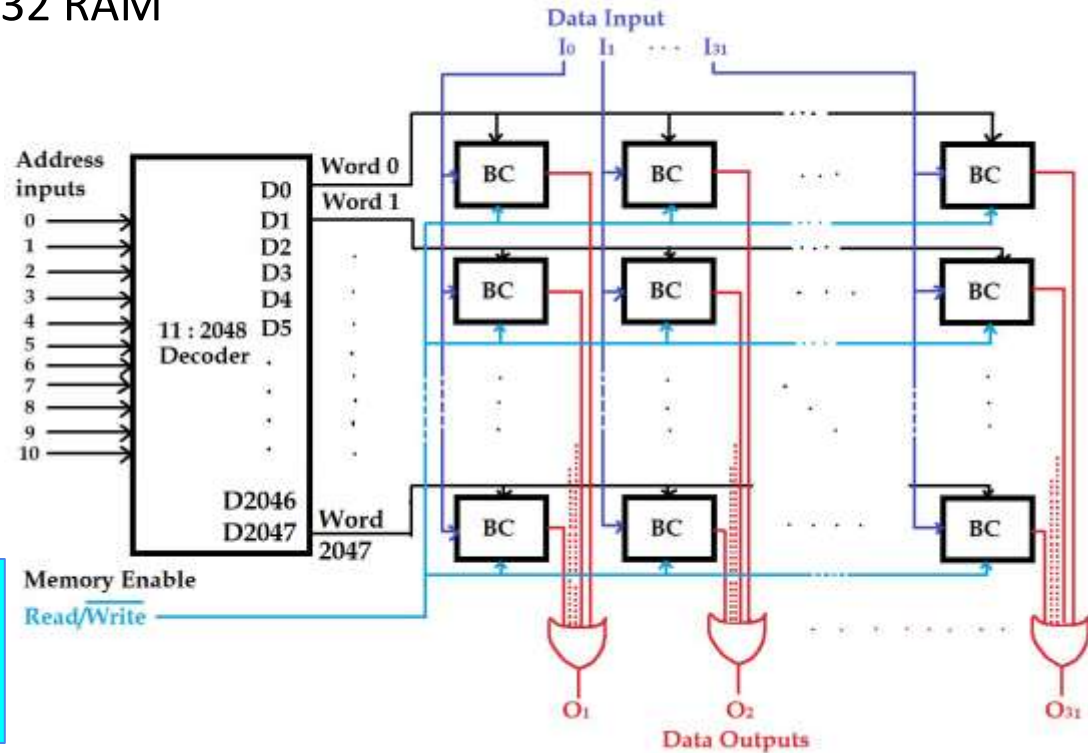
# of word

Bit/word

Logic construction of a 2048 x 32 RAM  
(with decoder and OR gates):

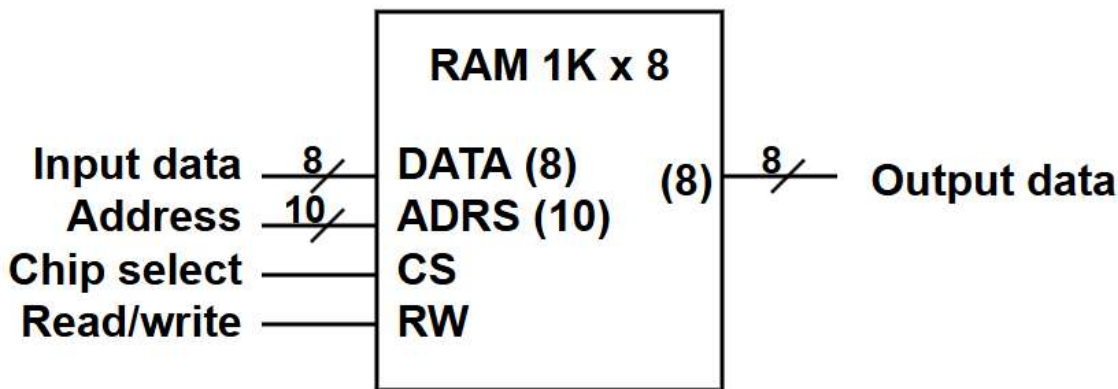


Since no of word is 2048, that would be equivalent to the decoders output. Since, the decoder will have 2048 outputs, so we should use a 11x2048 decoder



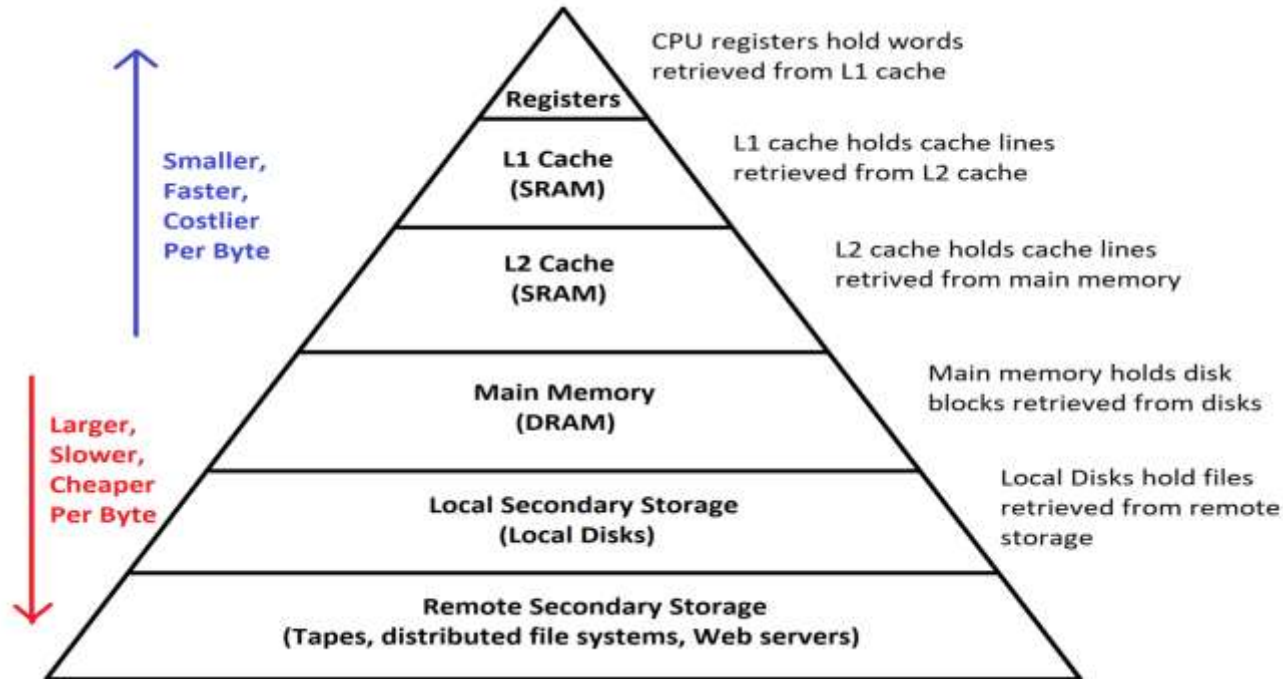
# Random Access Memory (RAM)

- An array of RAM chips: memory chips are combined to form larger memory.
- A 1K x 8-bit RAM chip:



Block diagram of a 1K x 8 RAM chip

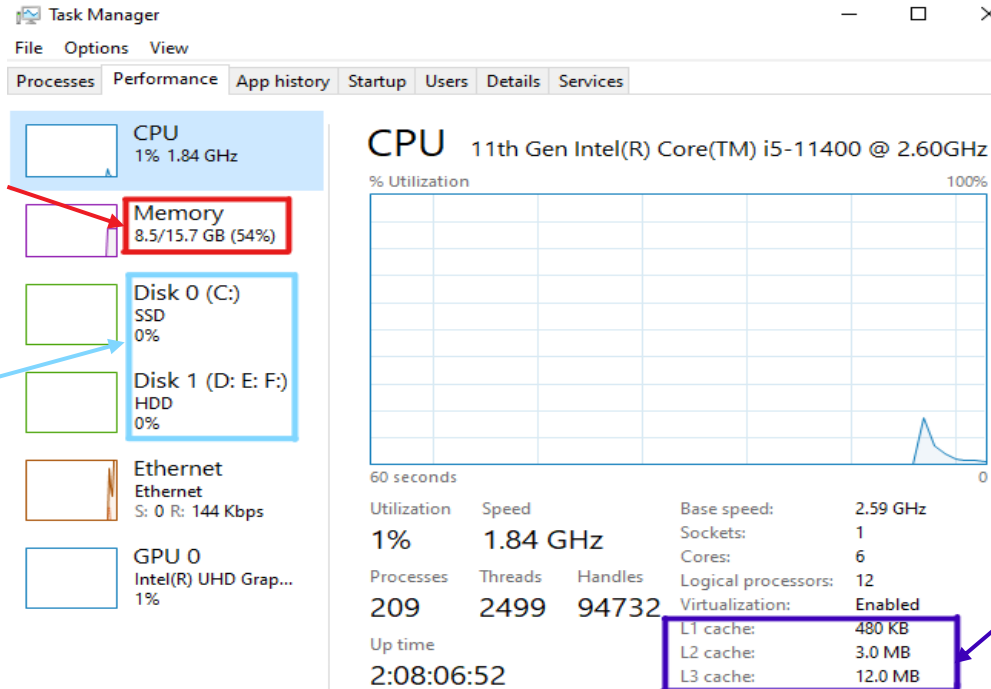
# Hierarchy of Memory [Self Study-not important for CSE260 exam]



# Check your PC's RAM Specifications From Task Manager (Windows) [Self Study-not important for CSE260 exam]

Main Memory:  
Dynamic RAM

Local Secondary  
Storage



Cache Memory  
Static RAM